

Microcircuits Special Feature

Microcircuits in action – from CPGs to neocortex

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To understand the interface between global brain function and molecular neuroscience – that is, the microcircuit level – a major challenge. Such understanding is prerequisite if we are to account for neural function in cellular terms. Very few vertebrate microcircuits are yet understood because their analysis is demanding technically. In this review of the *TINS Microcircuits Special Feature*, we attempt to shed light on the problem by comparing the operation of four types of microcircuit, to identify common molecular and cellular components. Central pattern generator (CPG) networks underlying rhythmic movements and hippocampal microcircuits that generate gamma and theta rhythms are compared with the neocortical microcircuits used in cognitive tasks and a cerebellar network. The long-term goal is to identify the components of a molecular and synaptic tool kit for the design of different microcircuits.

Introduction

Recent progress in neuroscience has been phenomenal on two extreme levels: the molecular biology of neurons and synapses on the one hand, and brain-imaging techniques on the other. The latter have provided important new knowledge about which parts of the brain are involved in different cognitive, emotional and motor functions. The maximal resolution for functional maps is the collective activity of thousands of neurons, whereas for neuronal and synaptic function it is the activity of individual molecules. The vast conceptual gap between these two levels is wide, bridged by only few and feeble strands. To understand the progressive emergence of complex function from the molecular level, it is essential to bridge this gap. Fortunately, the brain is modular: it consists of various networks that serve different functions in different brain regions, from the forebrain to the spinal cord. These functional modules, whether cortical columns or networks serving motor functions, form the interface between the molecular and higher-brain levels. One major

challenge of current neuroscience is to unravel the designs and modes of operation of these modules, which we refer to as ‘microcircuits’*.

The reason why until recently so little has been learnt about the dynamic processing that goes on in microcircuits, such as those of the neocortex and hippocampus, is that it has been technically demanding to study microcircuits systematically. We need to be able to identify the main types of neurons involved, draw a detailed circuit diagram, determine the forms of transmission and plasticity between different neurons, and study the dynamics of the microcircuit at cellular and synaptic resolution. Simulations are ultimately essential to explore whether a given set of data actually can account for a given function (De Schutter et al., in this issue). A multitude of demanding techniques have to be combined to address these issues, such as simultaneous intracellular recording from two or more interacting nerve cells. These techniques have now been successfully applied to several vertebrate circuits – the topic of this review – and previously also to certain invertebrate ‘simple systems’ circuits in crustaceans and molluscs [1–4].

Although the neurons, synapses and circuit diagrams of different vertebrate microcircuits can differ considerably, it is remarkable that all microcircuits identified so far in the brain display rhythmic behaviours, each at different preferred frequencies. In this *TINS Microcircuits Special Feature*, we explore the molecular, cellular, synaptic and circuit designs for the different forms of rhythmic behaviour observed in these microcircuits. This is crucial in a more general framework because it enables us to bridge the gulf from the single-cell level to integrated functions of the whole nervous system.

Rhythmic central pattern generator circuits

The microcircuits in the brainstem and spinal cord that generate respiratory and locomotor activity have been analyzed in considerable detail in different animal models,

* In this *TINS* special feature, the term ‘microcircuit’ is used to denote a minimal number of interacting neurons that can collectively produce a functional output, such as locomotor central pattern generators, hippocampal circuits producing gamma and theta rhythms, and circuits in the neocortex and cerebellum.

from the lamprey to mammals [5–11]. The general organization of respiratory and locomotor microcircuits has been investigated utilizing a battery of pharmacological, neurophysiological and computational techniques. The core of both networks is formed by pools of excitatory interneurons that generate burst activity and excite each other; inhibitory glycinergic or GABAergic neurons are not required in this process [12,13]. There is thus a core of one or many unit burst generators (unit central pattern generators, or unit CPGs) that can operate without inhibitory mechanisms. However, inhibitory mechanisms are crucial for coordinating the different segmental unit CPGs that generate left–right alternation in the lamprey spinal cord or alternation between hindlimbs in walking, and for coordinating flexors and extensors within a limb. Inhibition has a similar role in the respiratory system.

Figure 1 shows a simplified diagram of lamprey CPG organization. The network is activated by excitatory drive signals via reticulospinal neurons that excite both excitatory and inhibitory interneurons and the output motoneurons (for review, see [7]). They provide a command drive signal that turns on the network and controls the level of activity (burst rate 0.2–10.0 Hz). In each segment there is a pool of ~40–50 excitatory interneurons. The connectivity ratio between these excitatory interneurons has been estimated to ~10%, which means that when they become activated, further excitation will be generated

within the pool. Both the descending command and the excitatory interneurons release glutamate to activate both NMDA and AMPA receptors, and in addition gap junctions provide further excitation at many synaptic sites (Figure 1). The segmental excitatory interneurons receive additional excitation from the same type of neurons in nearby segments. The excitatory postsynaptic potentials (EPSPs) vary in amplitude at ~1 mV [7,14].

Interaction within the pool of excitatory interneurons will thus provide continuous depolarization to a degree sufficient to generate spike activity in a proportion of the cells, provided that the background excitatory drive can make some of the neurons generate action potentials. They will in turn excite motoneurons and also inhibitory interneurons with axons that inhibit the contralateral CPG. When one hemisegment is active, it will also inhibit the contralateral network, and when this inhibition ceases the other side will take over, owing to the background excitatory drive from the brainstem.

The excitatory interneurons thus interact to promote a burst, but how does such activity terminate in a predictable way? The main cause is spike-train adaptation, resulting from activation of Ca^{2+} -dependent K^{+} channels (K_{Ca}), which generates the post-spike after-hyperpolarization (sAHP), which is the main determinant of firing frequency. The sAHPs are summated, which results in a frequency adaptation (Figure 1f) that is

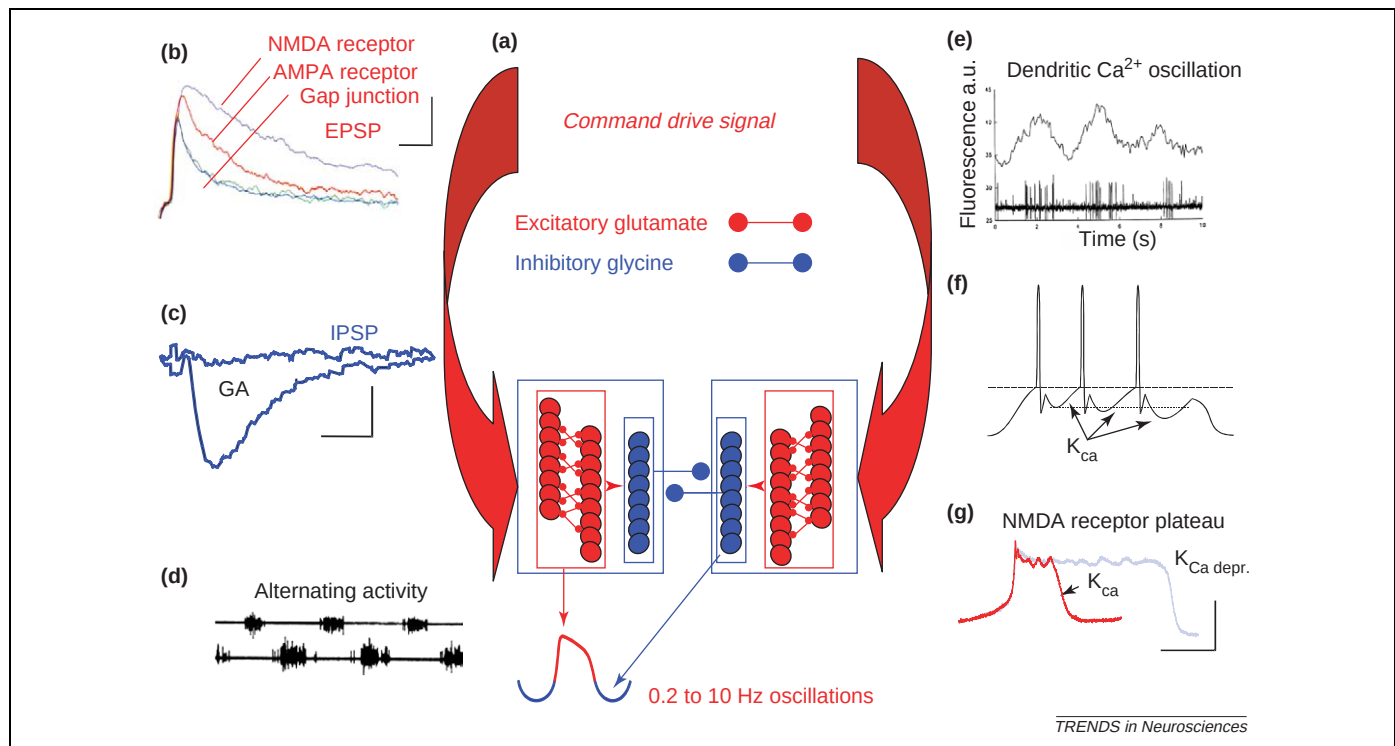


Figure 1. Spinal networks for locomotion and some of their electrophysiological characteristics. (a) The spinal networks that coordinate locomotion are located at the segmental level, and they each consist of a pool of excitatory glutamatergic interneurons (red) that provide excitation within the pool of interneurons. They are activated from the locomotor command centres in the mesencephalon and diencephalon via reticulospinal glutamatergic axons (large red arrows). The descending axons provide an EPSP that has three components, mediated by AMPA receptors, NMDA receptors and gap junctions (b). The glutamatergic interneurons in turn activate glycinergic interneurons (blue). These inhibit neurons on the contralateral side of the nervous system, illustrated here by their ability to generate inhibitory postsynaptic currents that are inhibited by glycine antagonists (GA) (c). Excitatory interneurons project to ipsilateral interneurons that receive out-of-phase inhibition from the contralateral side. This results in alternating efferent activity between the left and right sides (d). Of major importance for terminating burst activity is Ca^{2+} entry into the soma and dendrites during activity (e), which results in the activation of Ca^{2+} -dependent K^{+} channels (K_{Ca}). The action potential is followed by an sAHP due to K_{Ca} . The summation of the sAHPs leads to frequency adaptation (f), and the K_{Ca} also contributes to the termination of NMDA-receptor-mediated plateau potentials (g). Grey trace shows that the plateau is prolonged during depression of K_{Ca} current (K_{Ca} depr.). Time and voltage scale bars represent 5 ms and 0.5 mV in (b); 20 ms and 1 mV in (c); 20 mV and 1 s in (f).

greater the larger the sAHPs. In addition, Ca^{2+} entry through NMDA channels leads to a general activation of K_{Ca} channels, which promotes burst termination [15]. If the pool of excitatory interneurons in a hemisegment is pharmacologically isolated, it will still generate burst activity, but now at a fairly high burst frequency, from 2.5 Hz to >10 Hz. At lower frequencies, depolarizing plateaus due to activation of voltage-dependent NMDA receptors (Figure 1g) contribute to the burst activity [12,13]. These plateau depolarizations are terminated when K_{Ca} channels have been sufficiently activated, leading to closure of voltage-dependent NMDA receptors. Normal burst generation (with inhibition intact) will generate the entire range of locomotor frequencies from 0.2 to 10 Hz, and successive blockade of the inhibitory mechanisms not only blocks the alternation but also markedly increases the burst frequency. The inhibition is thus an important part of the burst-frequency control. In addition to this core circuit, there are also segmental inhibitory interneurons with ipsilateral action that possibly contribute to burst termination under some conditions, but they appear not to be crucial because their action can be blocked without any effects on burst generation. There are also a number of different types of modulation via G-protein-activated receptors [7,16], through neurons releasing glutamate, GABA, 5-hydroxytryptamine (5-HT), dopamine and neuropeptides (LeBeau *et al.*, in this issue).

Recently, it has become clear that excitatory core networks have a central role in other motor-related networks, such as those controlling respiration and saccadic eye movements. In the latter, microcircuits generate a brief intense burst to trigger the tectum (superior colliculus) to activate brainstem circuits and finally motoneurons to elicit the saccades [6,8,17,18]. Thus, excitatory circuits can be considered a general design feature for burst generation. However, an important requirement is that the bursts can be terminated in a controllable way. Membrane properties of microcircuit neurons, and in particular the presence of Ca^{2+} -dependent K^{+} channels, are important in this context (Toledo-Rodriguez *et al.*, in this issue).

Generation of synchronized network activity in the hippocampus

Another form of rhythmic activity is that generated in the brain, which can be recorded in the form of an electroencephalogram (EEG). Similar rhythmic activity can be recorded in the hippocampus both *in vivo* and *in vitro* and includes theta (4–12 Hz) and gamma (20–80 Hz) frequency oscillations. Gamma oscillations are thought to provide the temporal structure to coordinate the activity of many neurons, and gamma-frequency oscillation activity in the hippocampus has been implicated in encoding and retrieval of memory [19,20]. The theta frequency has been associated particularly with place representation and memory functions [21].

Gamma-frequency activity depends on networks of pyramidal neurons and interconnected GABAergic interneurons [20,22,23]. GABAergic inputs can time the firing of pyramidal cells generating network activities at theta and gamma frequencies. Single hippocampal inhibitory

cells innervate hundreds of pyramidal cells [24], so the output of a single interneuron can produce periodic inhibition entraining and synchronizing the firing of spontaneously active pyramidal cells [25].

Pyramidal cells form 80–90% of the total neuronal population in the hippocampus. In addition, 16 readily distinguishable subtypes of inhibitory interneurons have been identified [26]. Based on where their axons innervate pyramidal cells, two broad groups of interneurons have been identified (Figure 2): those whose axons target the perisomatic and proximal dendritic regions (e.g. basket, axo-axonic, bistratified and trilaminar cells) and those whose axons target more distal dendritic regions [e.g. trilaminar, oriens-lacunosum moleculare (O-LM) [26,27] and bistratified cells]. These two broad classes of interneurons exhibit important differences in their firing properties and contribute differently to gamma and theta activity.

Oscillations at gamma frequency

Interneurons are essential for the generation of gamma-frequency activity [20,22,23]. Fast-spiking interneurons, such as basket cells or axo-axonic cells, fire phase-locked to the gamma cycle and are capable of firing on every cycle of the oscillation [28–30]. They exhibit an amplified response to gamma-frequency inputs [31]. In addition, trilaminar cells, parvalbumin-positive basket cells and bistratified cells fire either doublets or single spikes on each cycle of the gamma oscillation, whereas O-LM cells fire at theta frequency [32,33] (Figure 2). Basket cells and bistratified cells have axonal arborizations restricted locally, whereas trilaminar cells [28] have projections extending across the hippocampus and possibly into the subiculum. Basket and bistratified cells might be involved in generating more local gamma signals, whereas trilaminar cells might propagate the gamma activity to more distal sites [32].

Electrical signalling via gap junctions

Electrical coupling occurs between pairs of interneurons usually of the same class in the hippocampus [32,34]. Such electrotonic coupling has been proposed to share excitation across the interneuron network [35]. Basket cell dendrites form a dense plexus, with individual basket cell dendrites often extending 500–800 μm [24] and interconnected by numerous gap junctions.

Gamma-frequency activity [36,37] is reduced or abolished by compounds known to block gap junctions. The importance of interneuron electrical coupling was investigated further using mice that lack the specific neuronal gap-junction-forming protein connexin 36 (Cx36). In adult mice, Cx36 is predominantly expressed in interneurons. Gamma-frequency oscillations were reduced in amplitude and synchrony [34,38] in these animals. However, the gamma activity in the Cx36-knockout mice was impaired but not fully abolished. Recently, a gene family encoding different gap-junction-forming proteins, the pannexins, has been identified [39]. They are expressed in the pyramidal cell layer in the hippocampus [40], but their role remains to be determined. Modelling studies have predicted coupling between pyramidal cell axons [41,42],

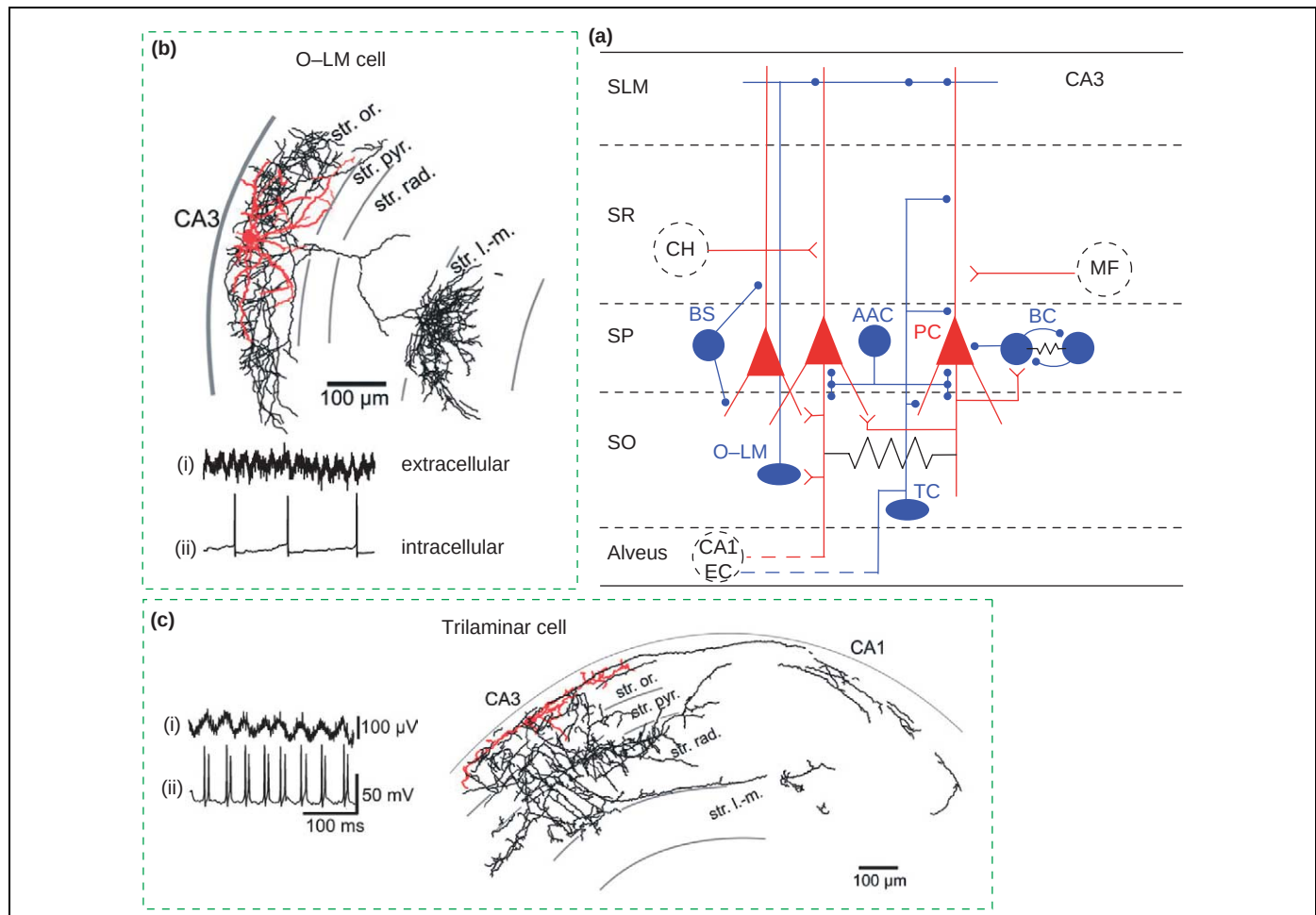


Figure 2. The hippocampal microcircuit. (a) Major cell types and synaptic connections in hippocampal CA3 region. Excitatory neurons are in blue, inhibitory neurons are in red, excitatory synapses are shown as V-shapes, inhibitory synapses are shown as circles, and electrical synapses are shown as a black zigzag. Dashed circles depict afferent and efferent extracortical input. Inhibitory synapses onto pyramidal cells (PC) are displayed according to the target domains: axonal inhibition is provided by axo-axonic cells (AAC), and somatic inhibition by basket cells (BC). Bistratified cells (BS), trilaminar cells (TC), and oriens-lacunosum moleculare cells (O-LM) target dendritic regions. Insets show morphological reconstructions of biocytin-filled O-LM (b) and trilaminar (c) cells in the stratum oriens (SO, str. or.). The soma and dendrites are in red and the axons are in black. The horizontal dendritic branches are restricted to stratum oriens for both cell types. However, the O-LM cell axons cross the pyramidal cell layer and arborize in stratum lacunosum moleculare (SLM, str. l.-m.). The axons of the trilaminar cell arborize in stratum oriens, stratum pyramidale (SP, str. pyr.) and stratum radiatum (SR, str. rad.) but do not extend into stratum lacunosum moleculare. Axon collaterals of the trilaminar neuron also extend into CA1 and reach the subiculum (not shown). Traces show extracellular field (i) and concomitant current-clamp (ii) recordings following induction of gamma-frequency activity evoked by pressure ejection of kainate. During gamma-frequency activity, O-LM cells fire intermittently at theta frequency, whereas trilaminar cells generate doublet action potentials on each cycle of the gamma activity. Panels (b) and (c) modified from [33]. Additional abbreviations: CH, contralateral hemisphere; EC, entorhinal cortex; MF, mossy fibre.

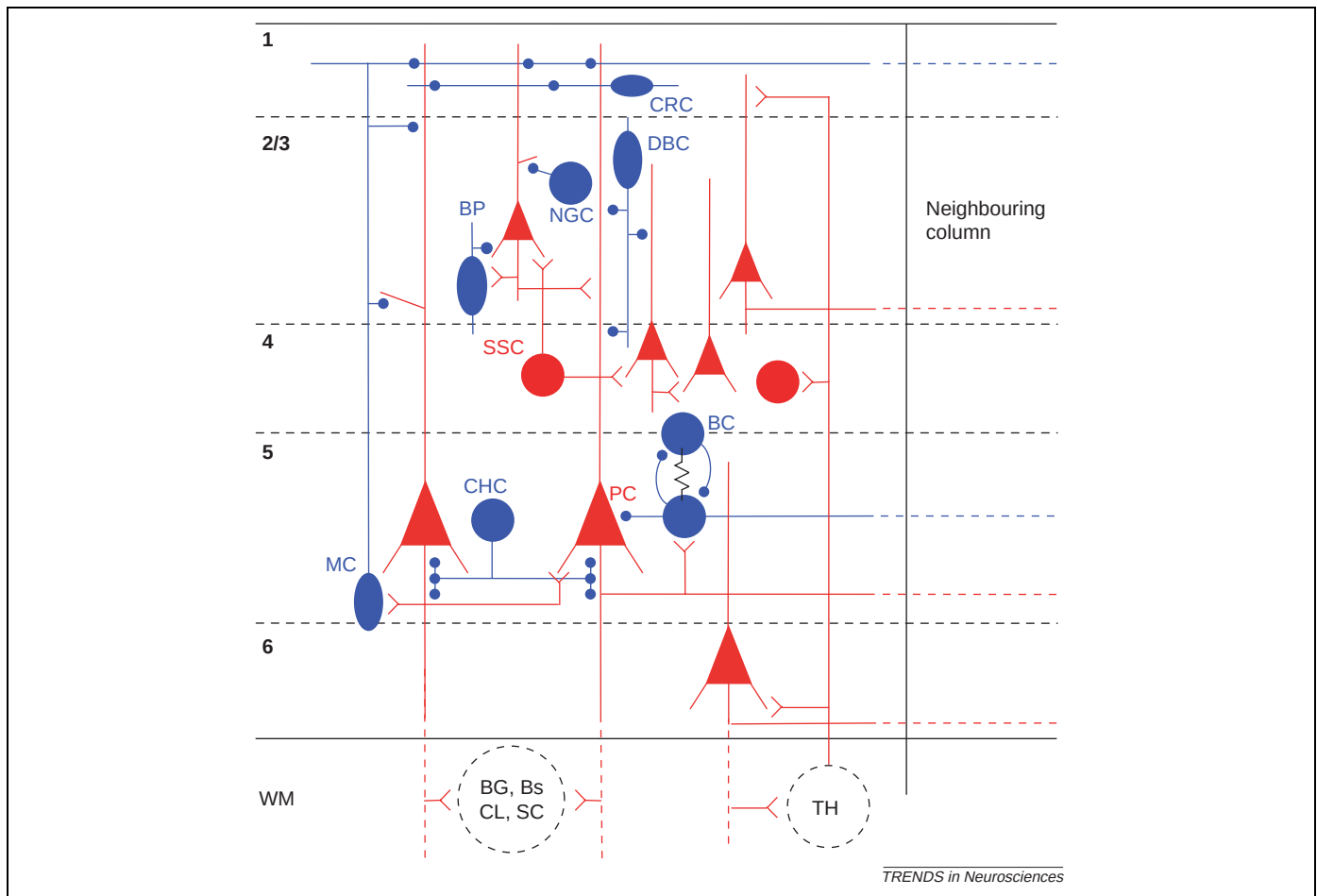
which could provide an essential excitatory driving force for the generation of gamma activity [43].

Oscillations at theta frequency

Theta activity was originally thought to depend entirely on inputs from the medial septum because inactivation of this region abolished theta oscillations [44]. However, they can be generated in hippocampal slices [45], indicating that the intrinsic hippocampal circuitry can generate these oscillations. The mechanisms that underlie the generation of theta activity are not fully resolved but, like gamma activity, theta activity is an inhibition-based rhythm in that inhibitory interneurons are essential [21].

Theta activity is associated with activation of many different subtypes of inhibitory interneurons [21]. Gamma-frequency and theta-frequency activity often occur together, and during theta oscillations basket cells also fire at gamma frequency, providing perisomatic inhibition [20]. However, unlike gamma oscillations,

theta oscillations have their largest amplitude in distal dendritic regions of pyramidal cells, and inhibition to these regions is crucial for theta activity. One class of interneuron of particular interest is that of the O-LM cells [26]. These dendrite-targeting interneurons fire at theta frequencies (Figure 2), a property related to their expression of the hyperpolarization-activated mixed cation current I_h [46]. Some interneuron subtypes fire at theta frequencies when depolarized by direct current [46] or when treated with agonists of ACh [47] or metabotropic glutamate [48] receptors. Recently, the firing properties of several morphologically different interneuron subtypes during theta oscillations *in vivo* have been characterized [49,50]. Firing of O-LM and bistratified cells coincided with the strongest hyperpolarization in the distal dendrites of pyramidal cells. By contrast, the basket cells and axo-axonic cells fired when the gamma power was highest and the pyramidal cell firing probability was lowest.



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Figure 3. The neocortical microcircuit – major cell types and synaptic connections. Excitatory neurons are in red, inhibitory neurons are in blue, excitatory synapses are shown as V-shapes, inhibitory synapses are shown as circles, and electrical synapses are shown as a black zigzag. Dashed circles depict afferent and efferent extracortical brain regions. Inhibitory synapses onto pyramidal neurons (PC) are displayed according to the target domains: axonal inhibition is provided by chandelier cells (CHC), somatic inhibition by basket cells (BC), and dendritic inhibition by double-bouquet cells (DBC), bipolar cells (BP), neurogliaform cells (NGC), Martinotti cells (MC) and Cajal-Retzius cells (CRC). PCs projecting to different brain areas reside in different layers: layer 5 is the main projection layer, with PCs projecting to subcortical regions such as the brainstem (Bs), spinal cord (SC), superior colliculus, basal ganglia (BG) and thalamus (TH). Layer 6 PCs project mainly to the thalamus, and PCs in superficial layers project to other cortical targets, such as neighbouring columns and the contralateral cortical hemisphere (CL). The representation of the different interneurons also changes across layers [44], with NGCs and DBCs mainly located in superficial layers, and MCs dominating the deep layers. BCs of different types constitute ~50% of interneurons in layers 2–6. Interneurons display diverse interlaminar targeting preferences: DBCs target dendrites that are typically located deeper than the soma, and MCs mainly target dendrites in the more superficial layers. BCs, NGCs, BPs and CRCs innervate neurons mainly within the same layer, although BC axons also spread laterally and innervate neurons from neighbouring cortical columns. Additional abbreviations: SSC, spiny stellate cell; WM, white matter.

The intrinsic membrane properties of hippocampal interneurons are important for determining their contribution to different types of oscillatory activity. In addition to the resonant properties of some interneurons when firing at gamma frequencies discussed previously [31], horizontal interneurons and pyramidal cells resonate at theta frequencies. O-LM cells show regular spiking and substantial accommodation, similar to pyramidal cells [51]. These intrinsic oscillatory properties could have a role in maintaining theta activity. By contrast, the fast-spiking basket cells that are activated during gamma-frequency activity [30,33] can fire at several hundred hertz without accommodation. Expression of the Kv3 family of K⁺ channels is important for this fast spiking behaviour

In conclusion

Gamma and theta rhythms can be expressed at the same time. Gamma activity depends on the interaction of fast-spiking inhibitory neurons targeting the perisomatic

region of pyramidal cells. Some of the interneurons are coupled through dendritic gap junctions that synchronize their activity, and thereby also provide a synchronous inhibition of the pyramidal cells. They become activated on the rebound following the inhibition, and also provide mutual inhibition to each other. Pyramidal cells resonate at the slow theta frequency, and so do the inhibitory O-LM cells that target the distal pyramidal dendrites. The O-LM cells fire in the theta rhythm, which is related to their specific cellular properties.

Neocortical microcircuit dynamics

The six-layered neocortical microcircuit (Figure 3) resembles in many aspects that of the hippocampus (Figure 2) but is more complex. It has several layers of pyramidal neurons, which constitute 80–90% of the cells in the neocortex. They excite each other, and also various interneurons (Figure 3) that provide feedback inhibition onto distal dendrites (Martinotti cells), onto proximal dendrites (double-bouquet, bitufted, bipolar and

neurogliaform cells), onto somata (small, nest and large basket cells) or onto axon initial segments (chandelier cells) [52]. It is likely that feedback inhibition from each of these classes of interneuron will tend to impose a specific form and frequency of synchronous activity. Interneurons receive and make various synapses and are endowed with a rich diversity of electrical properties, suggesting that every interneuron could impose a unique pattern of activity on the microcircuit.

Multiple mechanisms for oscillations have been revealed in isolated neocortical slices. When GABA_A receptors are blocked in neocortical slices, the microcircuit begins to oscillate at 4–10 Hz [53] in some but not all neocortical areas. These oscillations seem to depend on activation of AMPA receptors, which cause a Ca²⁺ influx that activates Ca²⁺ release from intracellular stores and modulates K⁺ channels to impose a period of excitability on the neuron. However, these oscillations also depend on pyramidal neurons, particularly those of layer 5, which express abundant I_h channels, mostly in their distal dendrites. These channels impose a resonance frequency on these neurons of ~4–7 Hz [54], which is a similar range to the theta rhythm in the hippocampus. Spike-train adaptation, caused by Ca²⁺-activated K⁺ channels, also has a profound influence on the rate at which oscillatory current input can synchronize pyramidal neurons, and has been shown to set the preferred frequency for the network oscillations [55]. Oscillations <1 Hz seem to depend on different mechanisms such as the basic synaptic connectivity mediated by AMPA and kainate receptors [53], although a crucial role for depressing synapses between pyramidal neurons has also been found in simulations [56] (Silberberg et al., in this issue). More recently, the facilitating excitatory synapses onto interneurons have also been found to be important in generating highly controlled and very slow oscillations (<<0.1 Hz; O. Melamed, G. Silberberg, H. Markram and M. Tosdyks, unpublished).

Different mechanisms seem to be involved in generating high rates of synchronous activity. Above 20 Hz and into the high-gamma range up to 80 Hz, feedback inhibition is important to orchestrate the timing of spikes in target neurons. Neocortical interneurons are generally powerful because they deploy many synapses onto their target neurons and even a single interneuron can entrain ~200 pyramidal neurons, enabling the network to oscillate at higher frequencies. Electrical coupling of interneurons, particularly interneurons of the same class [57–59], could further enable different interneuron classes to act in a concerted manner, which might be important to achieve very high rates of synchronous activity. Pyramidal neurons, in particular rhythmically bursting and chattering cells, have also been implicated at these high rates [60,61]. ACh, probably by blocking accommodation, facilitates higher synchronization rates [62]. Neocortical synchronous activity can be >80 Hz and even up to 200 Hz, which has been called ‘ripple’ activity. Fast-spiking parvalbumin-positive basket cells seem likely candidates for synchronizing pyramidal neurons at such high rates [63], and there is some evidence that rates >200 Hz do not depend on inhibition [64].

In this account, we have focused on the contribution of neocortical microcircuits to the generation of oscillations, but various other activity states, including synchronous activity, waves, and up and down states [65,66] could also be considered. Although the neuronal circuitry in different areas of neocortex is similar, one should recall that they are involved in vastly different tasks related to perception, cognition, memory and control of movement. The computations performed by a given microcircuit depend on the dynamic input from thalamus and other cortical and subcortical areas, and the efferent connectivity will determine the contribution to behaviour. Unlike the CPG in the spinal cord, which is related to a specific behaviour (locomotion), neocortical activity states produced during interactions with other forebrain structures are related to a spectrum of behavioural and cognitive states [67–71].

Cerebellar cortex microcircuit

The cerebellar microcircuit (Figure 4) has been considered a feedforward structure [72,73] where afferent information is processed by the granular layer and transmitted by the parallel fibres to the output neuron, the Purkinje cell. An essential feature is the synaptic plasticity (long-term depression, or LTD) that can be induced in the parallel fibre synapses on Purkinje cells (Figure 4) by interaction with the massive climbing fibre input from a single olivary neuron. LTD is not included in this review of dynamic microcircuit function, although it has been correlated to certain forms of motor learning [73].

The role of parallel fibres is being reconsidered (Box 1) and the cerebellar cortex is now known to oscillate at various different frequencies [74]. The presence of oscillations should not be a surprise, because all cerebellar inhibitory neurons (Golgi, basket, stellate and Lugaro cells) make recurrent connections (Figure 4). Such inhibitory microcircuits have a resonance frequency that is determined by the average delay time required for one inhibitory neuron to inhibit a nearby inhibitory neuron [75]. In particular, in purely inhibitory networks the oscillation period at resonance is equal to about four times the average latency to the onset of the evoked postsynaptic currents. This rule, in combination with computer simulations, has been used to predict the frequency bands of the oscillations, which match experimentally observed ranges [74].

The functional correlation of oscillations is best supported for slow oscillations in the 5–30 Hz range, which have been demonstrated to be coherent with different forms of movement [76,77]. Oscillations in this range are predicted to originate from feedback inhibition by Golgi cells onto granule cells, which in turn activate these Golgi cells by their parallel fibres [78] (Figure 4). This will lead to synchronization of Golgi cells receiving common parallel fibre excitation, as observed experimentally [79,80]. In addition, neuronal endogenous oscillatory rhythms as observed in Lugaro neurons upon activation by 5-HT [81] could also contribute. Cerebellar oscillation frequencies of 180 Hz were described by Lord Adrian in 1935 [81] and have more recently been reported in transgenic mice [74]. Such oscillations could originate in the highly interconnected network made by molecular

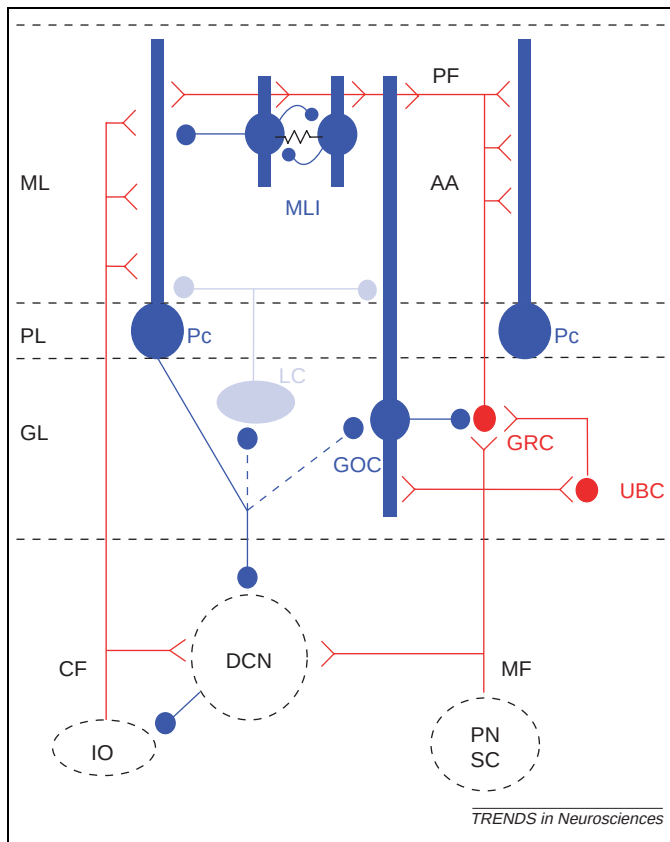


Figure 4. The cerebello-olivary circuit – the cerebellar cortex and its afferents. All neurons of cerebellar cortex are inhibitory except for the granule cells (GRC) and unipolar brush cells (UBC). The granule cell axon makes two kinds of synapses onto Purkinje cells (Pc): the ascending axon (AA) forms strong synaptic contacts that do not undergo LTD [83], whereas parallel fibres (PF) make weaker synapses that are sensitive to LTD [83]. A slow oscillatory circuit is formed by the feedback loop from excitatory GRC via PF to Golgi cells (GOC), which in turn inhibits the GRCs [74,78]. Both GRCs and GOCs are activated by mossy fibres (MF). Lugaro cells (LC) inhibit GOCs and Pcs and are intrinsic oscillators activated by 5-HT [81]. Fast oscillatory circuits are formed by the molecular layer (ML) interneuron network (MLI), which consists of stellate and basket cells recurrently coupled with inhibitory synapses and dendritic gap junctions, and by Pc–Pc collateral inhibition [74]. Another feedback loop is external to the cerebellar cortex and controls induction of LTD [73]. It is formed by Pcs inhibiting the deep cerebellar nuclei (DCN) and DCN inhibiting inferior olive neurons (IO) that have as their axons the climbing fibres (CF), which excite the Pc and promote induction of LTD at PF synapses. Additional abbreviations: GL, granular layer; PL, Purkinje layer; PN, pontine nuclei; SC, spinal cord.

layer inhibitory interneurons but, because of the high firing rate of Purkinje cells during these oscillations, it is likely that Purkinje cell axon collateral synapses also contribute [74,75].

Concluding remarks

What have we learnt from these four microcircuits, which all generate oscillatory activity but differ fundamentally in function? Do they have anything else in common? Yes, they do. The different CPGs (locomotion and respiration) have a core of excitatory neurons that generate recurring burst activity, overlapping with the theta rhythm in hippocampus and neocortex. In these two cortical areas, pyramidal neurons dominate (80–90%). They excite other pyramidal neurons located nearby. Pyramidal neurons resonate around theta frequency owing to their membrane properties, and in particular the sAHP in which Ca^{2+} -dependent K^{+} channels and I_h are important. The

Box 1. Granule cell output synapses are not all alike

Many theories on the function of cerebellar cortex have been inspired by the long parallel fibre tract and by the huge convergence of parallel fibre inputs onto the flat Purkinje cell dendritic tree [72]. The excitatory parallel fibres are formed by granule cell axons, which first ascend through the molecular layer and then bifurcate (Figure 4). A study of the properties of unitary granule–Purkinje synapses reported that only ~15% of the parallel fibre contacts generated measurable currents and suggested that most parallel fibre synapses are silent [84], presumably owing to synaptic LTD. Conversely, 50% of ascending axons made effective synapses with overlying Purkinje cells, confirming that these synapses could be more effective sources of excitation, as proposed in earlier *in vivo* studies [85]. The difference between the two synaptic populations could be due to different susceptibility to LTD, as was initially proposed based on differences in presynaptic morphology [86]. Recently, it was reported that indeed parallel fibre synapses are the only granule synapses in the cerebellum to undergo LTD. They have lower release probabilities than ascending-axon synapses, which are selectively resistant to LTD [83]. A summary of the new view on the cerebellar microcircuit is that ascending axons are essentially hardwired into the microcircuit and behave as event detectors, whereas parallel fibres have the capacity to undergo both short-term and long-term changes in synaptic strength and have a dynamic role in modulating cerebellar processing [87].

same factors are important for burst termination in the CPGs. There is thus a prominent excitatory interaction in the hippocampus and neocortex that is important for generating the oscillations, resembling the core microcircuits of the respiratory and locomotor CPGs [82]. The different burst generators within these CPGs are coordinated through inhibitory interaction, which generates the appropriate output patterns, for instance in terms of alternation between groups of antagonistic motoneurons.

The six-layered neocortical microcircuit (Figure 3) resembles that of the hippocampus (Figure 2) but is more complex. It has several layers of pyramidal neurons, each with different output targets (the brainstem, the spinal cord, the thalamus or other cortical neurons). Hippocampal activity is characterized by the theta and the much faster gamma rhythms. These rhythms are both due to pyramidal neurons that mutually excite each other, but they are also served by two different sets of interneurons. Those that inhibit the most distal dendrites are involved in the slow theta rhythm; they resonate at theta frequency and have a prominent I_h current, similar to that of pyramidal neurons. By contrast, fast-spiking interneurons such as inhibitory basket cells or axo-axonic cells target the soma and proximal dendrites; they discharge rapidly, have a brief afterhyperpolarization, fire phase-locked to the gamma cycle and are capable of firing on every cycle of the oscillation [28–30]. They fire in a synchronous fashion and are coupled by gap junctions, and they exhibit an amplified response to gamma-frequency inputs. Pyramidal neurons fire on the postinhibitory rebound, and thereby excite these fast-spiking interneurons. The crossed inhibitory neurons in the spinal CPGs also target the soma, and contribute through postinhibitory rebound to the next burst. The excitatory synapses are located distally, on the somatodendritic membrane. The circuit mechanisms underlying the different types of oscillatory activity appear to be similar in the hippocampus and

neocortex, although fewer details are known for the neocortex.

The cerebellar microcircuits are different because they rely on only two excitatory input channels: parallel fibre input from granule cells, which relays mossy fibre input, and climbing fibre input from olivary neurons. Except for granule and unipolar brush cells, all neuronal components in the cerebellar cortex are inhibitory. This includes the Purkinje output neurons with their extensive dendritic trees, which are subject to extensive synaptic input and plasticity.

In this review, we have provided examples of four types of microcircuit for which a wealth of new information has emerged. To some degree, there is now a tool kit of cellular and synaptic mechanisms that can be used in the nervous system to generate oscillatory circuits with different properties. We foresee a rapid expansion of knowledge at the microcircuit level to include different structures in the forebrain and brainstem. As we learn more about the microcircuit level, we will be able to bridge the gap between the molecular level and that of global neural function, and gain fundamental insights into how our nervous system works. In addition, we will most likely be able to identify a much richer repertoire of the tools used for precise circuit design in different parts of the nervous system.

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